VM networking modes (NAT, NAT Network, Host-only, Bridged): when to use each, and why vulnerable labs must never be bridged.

VM Networking Modes

- **NAT**: The VM sits behind a private virtual router managed by your host computer. The VM can access the internet, but external devices cannot talk to the VM.
    - *Use case*: Standard internet browsing or downloading basic updates.
- **NAT Network**: Multiple VMs share the same private internal network. They can talk to each other and access the external internet through a single gateway.
    - *Use case*: Setting up a multi-machine lab environment that still needs internet access.
- **Host-only**: Creates a completely closed network between your host computer and the VMs. There is no external internet access at all.
    - *Use case*: Analyzing dangerous malware or testing risky exploits safely.
- **Bridged**: The VM connects directly to your physical home router just like a real computer, getting its own IP address on your physical home network.
    - *Use case*: When you need other physical devices in your house to connect directly to the VM.

Lab Safety Warning

Never use Bridged mode for vulnerable target virtual machines (like Metasploitable or DVWA).

When you bridge a vulnerable lab machine, you expose its massive security flaws directly to your home network. Anyone on your local Wi-Fi—or any automated scanners hitting your home router from the internet—could compromise it. Once compromised, an attacker can use that lab machine as a pivoting point to hack your personal computer, phone, or smart home devices.